

Problem-Solution Fit

Date	11 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender – Loan Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Banks and Financial Institutions - Loan Processing Officers - Credit Verification Teams - Individuals Applying for Personal or Home Loans	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited processing time. - Large volume of loan applications. - Need to maintain data privacy. - Dependence on manual verification.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Loan Verification ✓ Easy to understand ✗ Slow process ✗ Human errors ✗ Difficult to scale Basic Digital Systems ✓ Electronic record management ✗ Limited prediction capability ✗ Requires manual decision making
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Manual loan verification requires significant time. - High number of applications during peak periods. - Possibility of inconsistent approval decisions. - Difficulty in assessing applicant eligibility accurately. Frequency: Daily	9. PROBLEM ROOT / CAUSE [RC] - Manual evaluation process. - Lack of intelligent decision-support tools. - Increasing number of loan requests. - Time-consuming verification procedures.	7. BEHAVIOR + ITS INTENSITY [BE] - Reviews multiple loan applications every day. - Cross-checks applicant information carefully. - Verifies financial documents before approval. - Frequently relies on experience for decision making. Intensity: High

3. TRIGGERS TO ACT [TR]

- Increase in loan applications.
- Need for faster approval decisions.
- Requirement to reduce manual workload.
- Demand for accurate risk assessment.

4. EMOTIONS BEFORE / AFTER [EM]

Before

- Stressed
- Uncertain
- Overloaded
- Concerned

After

- Confident
- Efficient
- Satisfied
- Productive

10. YOUR SOLUTION [SL]

The Smart Lender system applies Machine Learning algorithms to evaluate applicant information and predict loan approval eligibility. It analyzes financial and personal attributes, provides quick prediction results, reduces manual effort, improves consistency in decision-making, and assists bank officials in processing applications more efficiently.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Bank Web Portal
- Loan Management System
- Internal Banking Software

OFFLINE

- Branch Office
- Customer Documents
- Verification Meetings